

Status

September 15, 2026

CONTENTS

title: "Volatility Surface & Rough Heston Pipeline" subtitle: "Progress report — Session G" author: "Akin Akandere" date: "15 September 2021"

Status

Your review

Positivity preservation (flag 3)

The weighting scheme for H

Time-stepper constraints

Target metrics over n runs

Your questions, measured

What $|\varphi| \leq 1$ means, and whether a violation is a calibration error

Does the checked Lewis contour solve the fBm model? Is it Itô-isometric?

Does it need to read the tail at the cut-off?

What this does to analytics on real market data

A plain CIR filter vs the filter built for the rough model

The computational efficiency of the Session E change

Session G

Real data

Model fidelity

Roughness, learned by the filter

Engineering

Corrections

Open flags

What needs you

Next

Appendix: health and sanity checks

title: “Volatility Surface & Rough Heston Pipeline” subtitle: “Progress report — Session G” author: “Akin Akandere” date: “15 September 2026”

Status

Your review of the Sessions D–F report has been worked through, every question you asked has a measured answer, and today's four session items are done except the one that needs your hands: recording real IBKR data needs your login. Everything downstream of that login — recorder, pipeline, calibration, filters — is built and has been run end to end on a synthetic recording with a known answer.

	Sessions D–F (13 Sep)	now
Sessions complete	6 of 6	7 (G added)
Test suites	10	11, all under pytest
Test items / checks	— / 487	83 / 563
Analytical health checks	90 / 90	107 / 107
Production lines	6,475	9,123
Test lines	3,634	4,293
Commits	26	30
Remote	none	private GitHub backup (bredy00)

Several results in this report correct earlier statements, including two claims in the Session E report and one premise in your review. They are collected under *Corrections*.

Your review

The wins you listed stand as measured: the $\xi \rightarrow 0$ identity (2e-16), the Fornberg stencil (5.3e-12 on Breeden–Litzenberger) and the κ prior (spread 34.7% $\rightarrow$ 0.07%). Your remark about estimators built for fractional Brownian motion is picked up below: the roughness estimators are tested on exact fBm, and the time-series H estimator built today is checked against a planted truth the same way.

Positivity preservation (flag 3)

Built as your adapted QE scheme, on exact moments. In the lifted model the drift matrix $A = \text{diag}(x) + \kappa \mathbf{1} \mathbf{w}^T$ is similar to a symmetric matrix. In its eigen-coordinates every factor is a scalar OU process driven by the same $\sqrt{V}$ dB, so the conditional mean and covariance over a step are closed-form. The covariance is the Itô isometry evaluated exactly; it matches brute-force quadrature to 3.5e-12. Each step then does four things:

1. Draws V from Andersen's QE law with those two moments. V is never negative, and both moments are exact.
2. Assigns V 's surprise to the factors along the conditional regression direction.
3. Adds noise orthogonal to V , so $V = v_0 + c^T y$ holds exactly.
4. Draws that orthogonal noise jointly with the return's Brownian increment.

The joint draw in step 4 mattered. My first version drew it independently, and that lost the leverage carried into future variance by the slow factors: a +2.0e-3 error on a $k = +0.08$ call.

check	result
simulated rough variance, 3000 days ($\xi = 0.3$)	min $-2\text{e-}17$ (QE) vs below zero on 13% of days (Euler)
one step from $V_0 = 2\theta$, 200,000 paths	mean within 0.03 SE, variance -0.27%
252 steps from far out of equilibrium	mean within 0.5 SE, variance within 1%
prices at 250 steps, $\xi = 0.5$, three strikes	worst bias $2.7\text{e-}4$ vs Euler's $1.2\text{e-}3$; martingale

The Kalman filter's rough model now uses the same exact moments instead of the Euler transition. The QML κ bias, measured as one Newton step from the truth over 6 seeds, in SE:

data and filter	mean z
Euler data, Euler filter (Session F: the floor binds)	-2.63
QE data, exact filter, variance away from zero	+0.23 — unbiased
QE data, exact filter, variance at zero on ~13% of days	+1.79 (and -0.65 with smaller noise; spread ~2)

The floor's bias is gone. That does not remove every bias. Away from zero, the estimator is unbiased. Where simulated variance sits at zero on 13% of days, a Gaussian quasi-likelihood is unreliable in both directions, and no discretisation fixes that; it needs a non-Gaussian filter. This is recorded as a new, narrower flag.

The weighting scheme for H

One correction to the premise first. $\text{corr}(H, \xi) = 0.9776$ was measured with *equal* weights. Under vega² weights it is 0.99, and across every scheme and design studied it is 0.92–0.99. The correlation belongs to the surface design, and no reweighting removes it. What vega² weights destroy is the *standard error*: on the 20-quote surface they leave 2.3 effective quotes.

Both of your proposals are built (`calibrate/weights.py`):

- **Inverse variance:** each quote's own spread over its vega, plus a 0.1 vol-point model-error floor.

- **Hybrid:** inverse variance balanced so every expiry carries equal weight, plus one residual per expiry ≤ 30 days on the ATM skew. Each such residual is the local skew estimator applied to model and market vols alike, so the estimator's own bias cancels.

`study_h_weighting.py` measures each scheme at the truth by Gauss–Newton, then gives the market a surface the model gets wrong: once at the long end (a six-month hump plus extra long-dated curvature), once at the short end (1–3 day wings 2 vp rich).

RMS error in H, 90-quote recorder-like surface, realistic spreads:

scheme	SE(H)	long end wrong	short end wrong	worst case
vega ²	0.075	0.098	0.075	0.098
inverse variance	0.0076	0.026	0.087	0.087
hybrid (skew anchors)	0.012	0.028	0.055	0.055
hybrid, anchors $\times 25$	0.031	0.038	0.032	0.038

Inverse variance cuts SE(H) tenfold. When the long end is wrong it is the best scheme. When the short-dated wings are wrong it is the worst, because it trusts precise quotes there. The skew anchors fit the ATM *slope*, which a symmetric wing premium does not move, so the hybrid has the best worst case. The real-data pipeline uses the hybrid by default. Real surfaces are wrong somewhere, and we will not know in advance where.

Time-stepper constraints

The measured stability constants are now hard assertions at three levels.

1. **Precondition.** `log_char_func` refuses a mesh whose largest step exceeds the stable step for its $|u|$ range, before any work, with `StabilityError: ... needs  $\geq 383$  steps, got 120`. That is the $1.21e36$ runaway case.
2. **Postcondition.** `char_func` refuses $|\phi| > 1$ on $-1 \leq \text{Im } u \leq 0$.
3. **After a fit.** `calibrate_rough_heston` re-prices every expiry from scratch at the fitted parameters and reports `ok = False` if any strip fails.

Only convergence studies that probe beyond the edge on purpose switch the guards off, by name. Inside calibration the Lewis pricer sizes its step count from the same constants, so the precondition cannot trip there by construction. If it ever did, that would be a bug, and it would stop the run rather than return a price. A parameter region that is merely hard (a fatter tail, a failed step-halving check) still goes through the existing refinement and failure penalty, and the optimiser steps back.

Target metrics over n runs

Trend rules now carry your three targets. The health-check history records the host, so timing is compared only against the same machine; a CI runner is not this laptop.

target	rule	current
rough objective evaluation	rolling mean over the last 10 runs ≤ 0.75 s, same host	0.61–0.67 s idle; best of 3 per run
recursive MLE vs offline MLE	per run < 1 SE; max over the last 20 runs < 2 SE	0.23 SE
kernel error, $N = 24$	< 0.01 on [1 day, 2 y] and [1 hour, 2 y]	0.008697

Two findings behind these rows:

- **Timing noise.** On this laptop the same code timed 0.61–0.67 s idle, 0.85 s with a background job and 1.05 s when thermally throttled. A single timing is not a trend input, so each run now records the best of three.

- **The kernel target only holds for $H \geq 0.08$.** $N = 24$ meets it at 0.97% there, but reaches 1.05% at $H = 0.05$ and 1.13% at $H = 0.02$, and calibration may go as low as 0.02. Calibration now checks the kernel error at the *fitted* H and polishes again with $N = 32$, which stays below 0.67% on the whole box.

Your questions, measured

What $|\varphi| \leq 1$ means, and whether a violation is a calibration error

$\varphi(u) = E[e^{iuX}]$ with $X = \log(S_T/F)$. On the Lewis contour $u = v - i/2$:

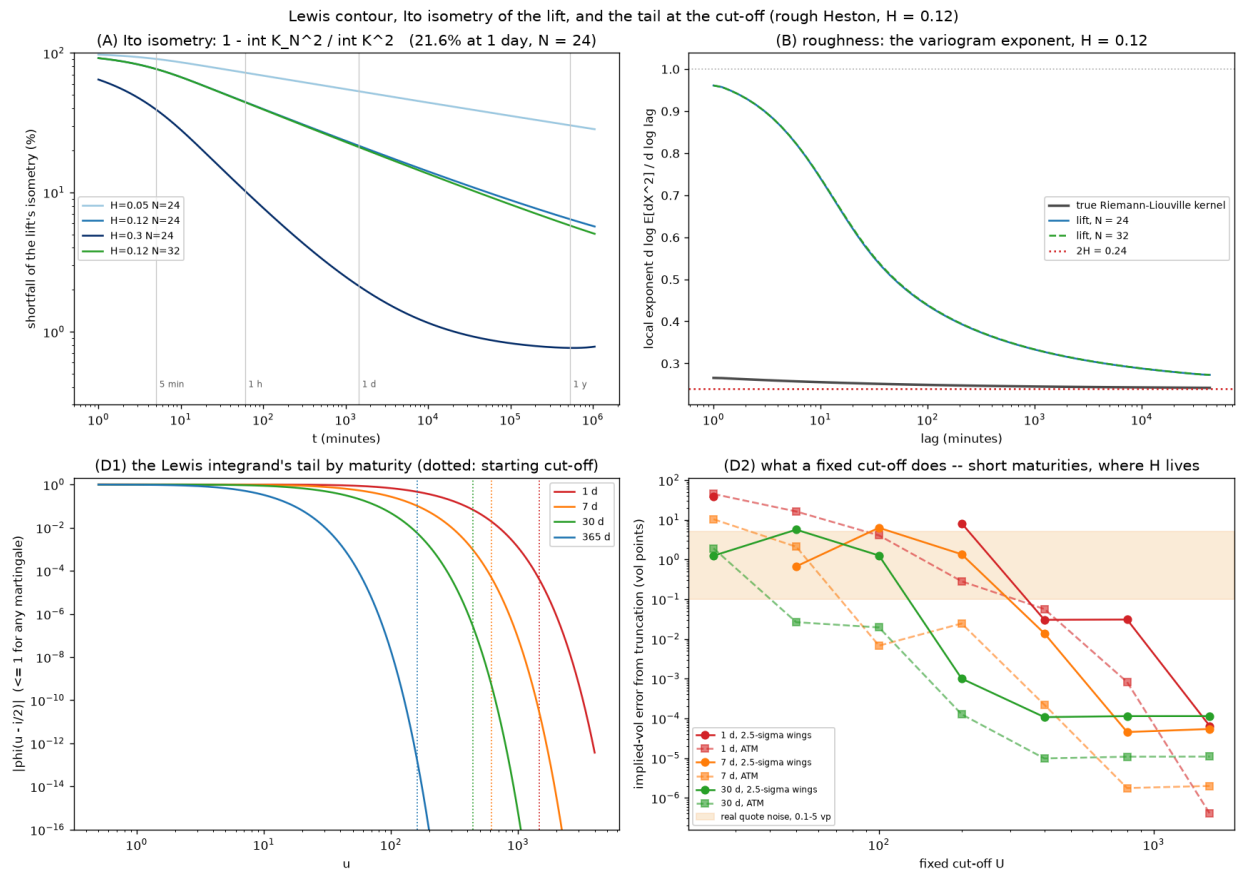
$$|\varphi(v - i/2)| \leq E[(S_T/F)^{1/2}] \leq (E[S_T/F])^{1/2} = 1$$

by Jensen's inequality and the martingale property. The same argument gives $|\varphi(u)| \leq 1$ on the whole strip $-1 \leq \text{Im } u \leq 0$. It holds for **every parameter set of every martingale model**: it is a property of a probability law.

So a computed $|\varphi| > 1$ is never a model value, and in that sense never a calibration error. It is always a broken solve. **But calibration is how a solve breaks.** The stable step depends on H , ξ , ρ , κ and on the u -range. Short maturities need a wide u -range, and an optimiser walking toward small H or large ξ crosses the edge of a step count that was fine at the start. That is why the constants are enforced *before* the solve.

The invariant also has a limit on real data. A wrong forward leaves the model's cf perfectly valid and shows up as skew and level misfit (ρ and H bias) instead. That is why the pipeline checks put-call parity and the τ convention separately.

Does the checked Lewis contour solve the fBm model? Is it Itô-isometric?



fBm. No, and it does not need to. Rough Heston's variance is driven by a Riemann–Liouville kernel against a Brownian motion, not by fractional Brownian motion. The characteristic function comes from the lifted Riccati system, and Lewis integrates it. fBm (Davies–Harte) appears only in the fixtures that test the roughness estimators.

Itô isometry — a finding. The lift is exactly isometric for its own kernel. Measured against the true kernel's isometry $\int_0^t K^2 = t^{2H} / (2H \Gamma(H+1/2)^2)$, the lift's variance of the Volterra driver falls short:

t	1 hour	1 day	1 month	1 year
H = 0.05, N = 24	72%	53%	38%	30%
H = 0.12, N = 24 (shipped default)	44%	21.6%	10%	6.4%
H = 0.12, N = 32	44%	21.1%	9.8%	5.8%
H = 0.30, N = 24	10%	2.1%	0.9%	0.8%

The sup-norm kernel error on the same band is 0.87%, and it cannot see this. $K^2 \propto s^{2H-1}$ ($s^{-0.76}$ at $H = 0.12$) puts 28% of the one-day isometry below seven minutes, where the lift's kernel is bounded, and 93% of the gap sits there. The rougher the kernel, the worse it gets. Raising N does not help (N = 40: 20.9%). Only a faster top node does, and that raises the sup-norm error unless N rises with it.

Does it move prices? To answer that I wrote the true rough Heston characteristic function with no lift: the fractional Adams scheme (Diethelm–Ford–Freed), used as El Euch & Rosenbaum do. It converges at second order to closed-form Heston at $H = 1/2$, and its own error here is $\leq 2.5e-5$ vol points. Against it:

maturity	1 d	7 d	30 d	90 d
H = 0.12, lift N = 24 (shipped): worst IV error	0.15 vp	0.06 vp	0.04 vp	0.03 vp
ATM skew error	-1.6%	-0.7%	-0.4%	-0.3%
H = 0.12, lift N = 40, top node 1e8	0.03 vp	0.03 vp	0.03 vp	0.02 vp
H = 0.05, lift N = 24: worst IV error	0.19 vp	0.08 vp	0.05 vp	—
ATM skew error	-1.5%	-0.7%	-0.5%	—
H = 0.05, lift N = 40, top node 1e8	0.04 vp	0.03 vp	0.03 vp	—

Prices integrate the kernel once more and largely wash the gap out, even at $H = 0.05$ where the isometry is 53% short at one day. The gap still matters when H is estimated from *intraday* variance lags, where the lift is effectively smoother: its variogram exponent over one hour to one month is 0.34 against $2H = 0.24$.

The drag is inside the cf. $E[X_T] = -i \varphi'(0)$ equals $-1/2 E[\int_0^T V dt]$ from the closed-form affine mean to $2.5e-8$ (1 day), $5.9e-7$ (30 days) and $3e-8$ (1 year); $\varphi(-i) = 1$ exactly.

Does it need to read the tail at the cut-off?

Yes. The $|\varphi| \leq 1$ bound gives a model-free truncation bound $e^{k/2} / (\pi U)$. It is useless: $U \approx 3 \cdot 10^5$ for $1e-6$ in price. Here is what a fixed cut-off does to 2.5-sigma wings:

fixed U	25	100	200	400	checked
1 day	39 vp	fails	8.2 vp	0.03 vp	U = 2410: $\sim 1e-4$ vp
7 days	fails	6.3 vp	1.4 vp	0.01 vp	U = 1113: $\sim 5e-5$ vp
30 days	1.3 vp	1.3 vp	0.001 vp	$1e-4$ vp	U = 531

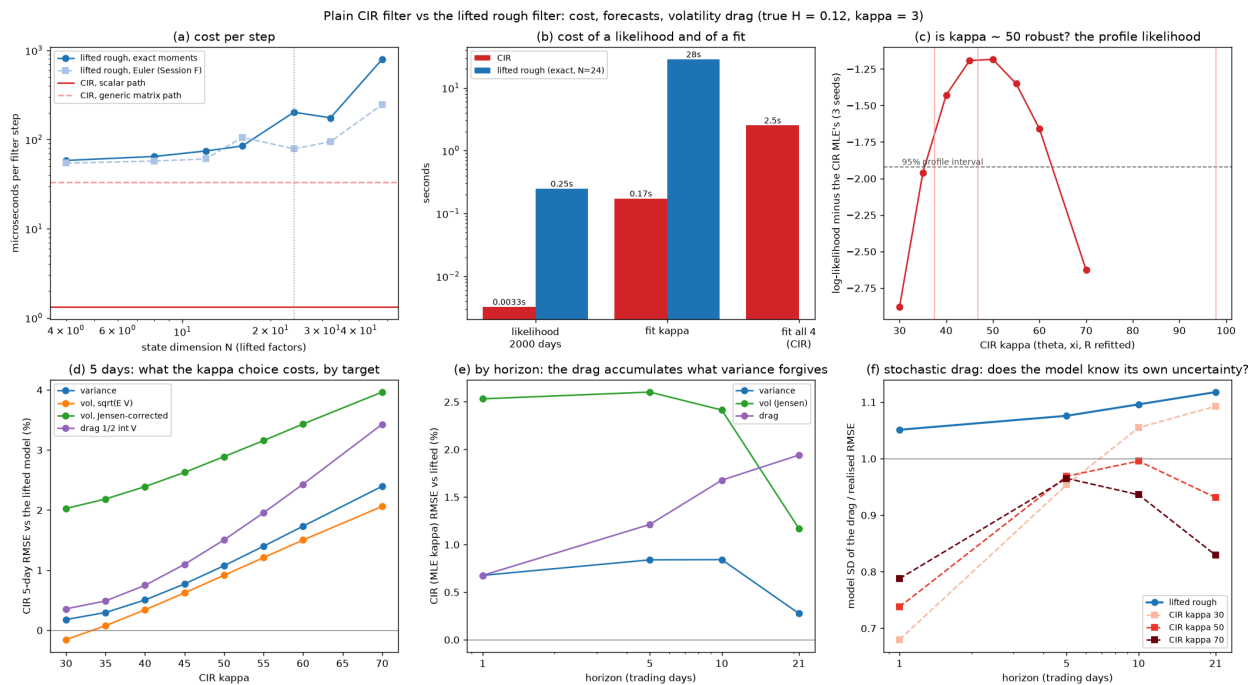
("fails" means the truncated price left the no-arbitrage band and no implied vol exists.) The errors are largest exactly where H is identified.

What this does to analytics on real market data

Measured or fixed today, in order of damage avoided:

1. **The clock.** τ was computed on a naive local clock against a UTC close that ignored daylight saving: about 2 hours short on this UTC+3 machine, 7% at one day. That biases the short-end term structure, hence H . Fixed and pinned.
2. **Vega units.** IBKR quotes vega per vol point; the code assumed per unit vol. Every IV error came out 100× too big, and the usability filter would have discarded almost the whole chain, wings first. Fixed; `record_chains.py --check` verifies the units on the first live quote.
3. **The tail at the cut-off,** above: 1–39 vol points at one to seven days if unchecked.
4. **The lift's fidelity:** 0.15–0.19 vp at one day, at $H = 0.12$ and 0.05. That is below real short-end quote noise (0.2–5 vp), so it is acceptable for calibration.
5. **The skew-slope estimate of H is biased by vol-of-vol.** A rough Heston surface with $H = 0.10$ and $\xi = 0.4$ gives a skew-slope “ H ” of -0.03 , as Session E found at $\xi = 0.5$. Never read H off the skew slope alone.
6. **Realised variance is an integral** (see Session G below). Treated as spot variance, it biases the filter's H from 0.10 to 0.39.
7. **Still open:** SPY options are American (the pipeline uses OTM quotes only, and the early-exercise premium there is small); calendar-time τ for 1–3 day options spans nights and weekends; IBKR's 30-day IV carries the variance risk premium, so it is not physical variance.

A plain CIR filter vs the filter built for the rough model



Cost. A CIR filter step costs 1.2–1.3 μ s. A step of the 24-factor rough filter with exact moments costs 120–200 μ s; it varies run to run on this laptop, and 79 μ s with Session F's Euler transition. A likelihood over 2,000 days takes 2.4–3.3 ms against 239–248 ms, and fitting κ takes 0.17 s against 19–29 s: about a hundred times the cost, for a state space that is right.

$\kappa \approx 50$ is not robust. On positivity-preserving data the CIR MLE of κ is 37, 47 and 98 on three seeds. The profile likelihood is nearly flat: within 1.92 log-likelihood units of the maximum from about 36 to 62, on average across seeds. κ is weakly identified.

Volatility drag. Over the sweep $\kappa \in \{30, \dots, 70\}$, CIR's error relative to the lifted model's:

horizon	variance	vol, $\sqrt{E[V]}$	vol, Jensen-corrected	drag $\frac{1}{2}\sqrt{V} dt$
1 day	+0.3 ... +1.3%	-0.0 ... +1.6%	+2.3 ... +3.2%	+0.3 ... +1.3%
5 days	+0.2 ... +2.4%	-0.2 ... +2.1%	+2.0 ... +4.0%	+0.4 ... +3.4%
21 days	-0.0 ... +0.6%	-1.5 ... -0.7%	+1.3 ... +1.6%	-0.3 ... +4.7%

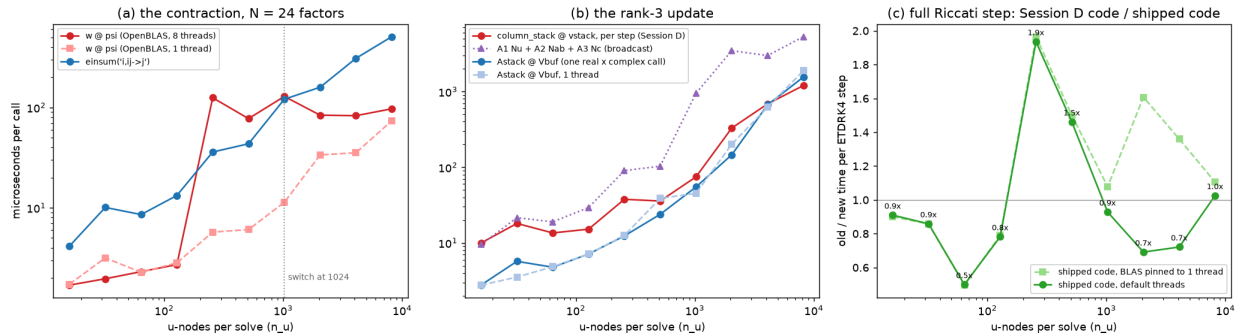
So the “3% at 5 days” survives the drag in the *mean*: expected drag is forecast almost as well. Low κ (30–40) is better than the MLE κ , and drag error grows with κ and horizon. Two places the imitation fails:

- **Volatility.** Once the Jensen correction uses each model's own variance of V , CIR is 2–4% worse at every horizon. Its point forecast is right; its *variance of variance* is not.
- **The stochastic drag.** At one day, CIR's forecast standard deviation of the drag is 0.68–0.79 of the realised one, and its 90% interval covers 85–89%. The lift: 1.05 and 93%.

For risk (VaR of a return, whose drag term is random) CIR is overconfident at short horizons at every κ tried.

The computational efficiency of the Session E change

Where the Riccati step's time goes -- lifted rough Heston, $N = 24$

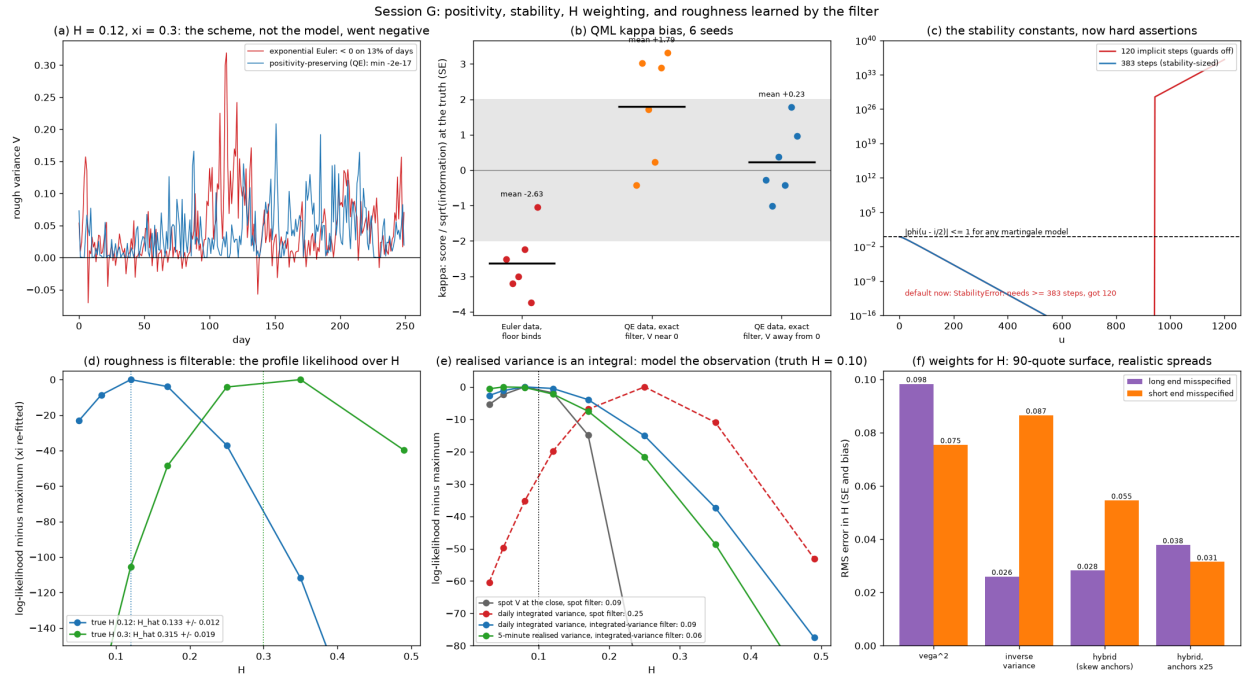


Measured on an idle machine, against the Session D loop exactly as committed (104f74d):

- **The contraction ($w \circ E$) @ ψ , $N = 24$.** OpenBLAS with 8 threads pays 80–130 μs per call once $n_u \geq 256$ (thread dispatch). einsum is 2–5 $\times$ slower than single-threaded matmul at small n_u . **Single-threaded matmul is fastest at every size:** 1.7 μs at $n_u = 16$, 11 μs at 1,024, 74 μs at 8,192.
- **The rank-3 update.** The pre-stacked product is 2–4 $\times$ faster than Session D's per-step `column_stack @ vstack` up to $n_u = 2,048$.
- **The full ETRK4 step:** 1.5–1.9 $\times$ faster at $n_u = 256$ –512, and 0.5–1.0 $\times$ elsewhere (timings on this laptop carry $\pm 30\%$ noise between runs).

The report's “6–13 $\times$ per step” and “1.3 ms per step starting threads” do not reproduce here. They may have been measured under CPU contention, which inflates thread dispatch; I cannot verify that after the fact. Calibration now pins BLAS to one thread, which removes the dispatch entirely. End to end, the gain on a 10-expiry objective is within timing noise on this laptop; the reason to keep it is robustness under load.

Session G



Real data

The recorder.

- `record_chains.py` records self-consistent cycles through the replay source:
 - a seed pass at both rights near the money gives this cycle's put-call-parity forwards;
 - the OTM grid is laid on a sigma band around those forwards;
 - one sweep fills it.
 - Cycle isolation:** request ids move on every cycle, so a late tick can never land on another contract. The control test shows it would.
 - Expiries** are chosen by target maturity (1 day to 1 year); on SPY's dailies the old "first n" rule stopped inside a month.
 - `--check` is a 15-second smoke test that verifies vega units on a live quote.
- `record_history.py` pulls five years of daily OHLC, IBKR's 30-day implied and historical vol, and twelve months of 5-minute bars, a week per request, inside IBKR's pacing limit.

Both are tested offline against `fake_ib.py`, a TWS stand-in that delivers callbacks on a reader thread with IBKR's tick types and units. Building it exposed four bugs that no stand-in feed could have shown: the clock, vega units, delayed greeks on tick 83, and a live-to-delayed fallback that never re-sent the spot request (latent since v2).

The pipeline, `run_real_data.py`:

- Recorded chain:** unit and convention checks, a hybrid-weighted surface with IVs from the mid under our τ and forward, then Heston and rough Heston fits with their post-fit checks.
- Recorded history:** realised variance, CIR and rough filters with H learned.
- Output:** four independent readings of H side by side — skew slope, surface calibration, filter profile likelihood, structure function.

`--synthetic` runs the identical pipeline on a fake-exchange recording priced by rough Heston, and on a 5-minute simulated history with $H = 0.10$. It recovers $H = 0.085 \pm 0.025$ from the history.

Not done: the recording itself. No TWS or IB Gateway is installed on this machine, and logging in needs your credentials and your phone. See *What needs you*.

Model fidelity

This is the positivity scheme above, now the default in the filter, the simulator and the real-data pipeline. Session F's rough results were measured under Euler, so their tests are pinned to Euler and labelled, and they reproduce exactly (CIR κ 48, 54, 43; -0.6% , $+2.8\%$, $+1.8\%$).

Roughness, learned by the filter

- **The profile likelihood over H**, with ξ (and optionally κ) re-fitted at every H.
 - True H = 0.12: **0.133 ± 0.012** . H = 0.49 is rejected by 266 log-likelihood units, although ξ was free to climb to 1.70.
 - True H = 0.30: 0.308 ± 0.015 .
 - With κ free as well, identification weakens but holds: 0.133 ± 0.019 and 0.303 ± 0.020 . κ trades against H, since fast mean reversion mimics roughness.
- **A bank of filters over an H grid**, giving an online posterior for H (0.121 ± 0.007 on the same data). A forgetting factor lets the posterior move if roughness changes.
- **A recursive MLE with H as a filter parameter** (`LiftedRoughModelH`).
 - For any H the lift keeps the same partition of the rate axis, so the state keeps its meaning while H moves.
 - The observation vector $w(H)$ moves with H, and the sensitivity equations now carry dH terms.
 - From H = 0.30 it reaches 0.120 within 100 days and ends at 0.119; the offline MLE is 0.115.

Realised variance is an integral. Daily realised variance is not a noisy reading of spot variance. It measures the variance integrated over the day, and integration smooths a rough path. On 500 days of synthetic rough variance with H = 0.10:

observation fed to the filter	H recovered
spot variance at the close + noise, spot filter	0.087 ± 0.017
true daily integrated variance, spot filter	0.25
realised variance from 5-minute bars, spot filter	0.39
true daily integrated variance, integrated-variance filter	0.092 ± 0.027
realised variance, integrated-variance filter	0.061 ± 0.027

`LiftedRoughRVModel` augments the state with the day's integrated variance. Its joint conditional moments with the factors are affine in the state, computed by graded Gauss–Legendre quadrature and checked against the closed-form factor covariance to $6.6e-16$ and against Monte Carlo to within 0.6%. Realised variance's own sampling noise, $2/M \cdot RV^2$, enters as a state-dependent observation variance. The real-data pipeline uses this model; fed real realised variance, the spot model would have reported $H \approx 0.4$.

Engineering

- **All eleven suites run under pytest.** A root `conftest.py` makes every `test_*` function a test item that fails if any of its checks failed, with the failing check names as the message. Nothing was rewritten, so running a suite as a script and under pytest cannot disagree.
- **Two tiers:** `pytest -m "not slow"` (~3 minutes) and `pytest` (everything, 11 minutes).
- **GitHub Actions** runs the quick tier on every push and the full tier plus health checks weekly and on demand. The health-check history is carried across CI runs in the Actions cache, so the n-run trend rules work in CI too.
- **The backup** is a private GitHub repository under bredy00, named `hurst` — the quantity everything here converges on. The IBKR recordings it will hold are licensed market data, so it is private until you decide otherwise.

Corrections

1. **corr(H, ξ) = 0.9776 was the equal-weight value, not vega².** vega² gives 0.99, and all schemes 0.92–0.99. vega² destroys SE(H), not the correlation.
2. **Session E: “6–13× faster per ODE step” and “1.3 ms per step starting threads”** do not reproduce on an idle machine: 1.5–1.9× at $n_u = 256$ –512, 0.5–1.0× elsewhere; dispatch is 80–130 μ s per call.
3. **Mid-session today I measured BLAS pinning at 0.717 $\rightarrow$ 0.641 s per objective.** It did not reproduce within timing noise; the pinning is kept for robustness, not speed.
4. **“A positivity-preserving scheme removes the QML bias”** holds away from zero only. Near zero the Gaussian quasi-likelihood is unreliable.
5. **Session F: “a CIR fitted to rough variance pushes κ to ~50”.** On positivity-preserving data, κ ranges 37–98 over seeds with a flat profile; the number is not robust.
6. **“Kernel error 0.87%” reads as “the lift reproduces the kernel”.** The variance of the Volterra driver is 21.6% short at one day. Prices are still within 0.15 vp of true rough Heston.
7. **The health check’s single timing of the rough objective** ranged 0.61–1.05 s on unchanged code; it is now the best of three.
8. **Session F’s rough-host filter results** were measured under the Euler scheme; those tests are now pinned to Euler and labelled as such.
9. **The live-to-delayed data fallback never re-sent the spot request**, carried unnoticed from v2.

Open flags

1. **Real data is not recorded.** It needs the IB Gateway login (tutorial below).
2. **Near-zero variance:** the Gaussian QML is unreliable when variance sits at zero; a particle filter or a characteristic-function likelihood is the fix.
3. **Realised variance's noise** is modelled for a diffusion. Microstructure noise and jumps inflate 5-minute RV; bipower variation is the robust alternative.
4. **Calendar-time τ for 1–3 day options** spans nights and weekends, which distorts the short end, where H lives.
5. **The lift's short-lag isometry gap** matters for intraday estimation of H. The finer lift ($N = 40$, top node $1e8$) fixes it for about $1.7\times$ the cf cost and $2.5\times$ the filter cost.
6. **The skew-slope H estimator is biased by vol-of-vol;** use it as a cross-check only.
7. **IBKR's 30-day IV carries the variance risk premium,** so it is not usable as a physical variance observation without modelling that premium.
8. **Rough host jump mode** (driver vs direct), carried from Session F: your decision.
9. **Hawkes on real events,** and the nearly-unstable Hawkes $\rightarrow$ rough volatility link.
10. `volatility_surface_3.py` **should be split.**

What needs you

1. Record the data. `docs/tutorial-ibkr-recording.md` walks through it step by step. In short: install IB Gateway, log in with API settings (read-only), then

- `record_chains.py --check` (15 seconds)
- `record_history.py` (any time, ~10 minutes)
- `record_chains.py --every 15 --until 15:45` during US market hours: 16:30–23:00 Istanbul until 31 October

Then tell me “recorded”, and `run_real_data.py` does the rest.

2. Four decisions.

1. Rough jump mode (carried from Session F).
2. Particle filter or cf-based likelihood for near-zero variance.
3. Whether to adopt the finer lift for intraday H work.
4. Whether the GitHub backup should stay private (recommended while it holds recorded market data).

Next

1. The first real-data run: the four readings of H on SPY, and the Heston vs rough Heston fit on a real surface.
2. A non-Gaussian filter for the zero boundary.
3. A trading-time clock for short-dated options, measured on the real chains.

Appendix: health and sanity checks

Foundations

check	measured	threshold	verdict
norm_pdf vs scipy	0	1.00e-15	PASS
norm_cdf vs scipy	2.22e-16	1.00e-15	PASS
put-call parity C-P = F-K	2.84e-14	1.00e-09	PASS
vega vs central difference	4.23e-09	1.00e-04	PASS
implied-vol round trip	8.05e-16	1.00e-06	PASS
fd_first exact on a quadratic	7.46e-14	1.00e-12	PASS
fd_second on a quadratic vs its rounding floor 2.08e-11 measured, floor 5.06e-11	0.41 x floor	1 x floor	PASS
shipped 2nd derivative: order on a \$1/\$5 kink grid Fornberg 5-point; mapped-coordinate idea measured at 0.01	3.077	2	PASS
3-point stencil order on the same grid (kept for contrast) the documented first-order loss	1.086	1.5	PASS

Heston cf

check	measured	threshold	verdict
$\phi(0) = 1$	0	1.00e-12	PASS
$\phi(-i) = 1$ (martingale)	0	1.00e-10	PASS
$\phi(-u) = \text{conj}(\phi(u))$	0	1.00e-12	PASS
$ \phi(u) \leq 1$	0.9999	1	PASS
branch continuity, shipped (max/median step) Albrecher $g=(a-d)/(a+d)$	5.242	50	PASS
branch discontinuity, reciprocal form the trap, kept to prove it is real	2825	500	PASS
trap / shipped separation	538.9 x	100 x	PASS
$\xi \rightarrow 0$ degenerates to Black-Scholes at $\xi=1e-8$, cancellation-free form; was 5.9e-09 at $\xi=1e-4$	2.22e-16	1.00e-14	PASS
naive-form cancellation at $\xi=1e-6$ (kept to prove it) shipped form is 5.7e-13 at the same ξ	3.44e-05	1.00e-06	PASS

Pricers

Status

check	measured	threshold	verdict
Lewis vs exact Black-76	3.55e-15	1.00e-10	PASS
Carr-Madan vs exact Black-76	1.94e-16	1.00e-10	PASS
Lewis vs Carr-Madan on Heston independent routes, 5 maturities	4.99e-15	1.00e-08	PASS
Carr-Madan insensitive to damping alpha alpha 0.75 to 3.0	6.11e-16	1.00e-08	PASS
Fourier vs Monte Carlo (sigmas) residual is Euler O(dt) bias	0.3618 sd	5 sd	PASS
simulated $E[S/F] = 1$	1.94e-06	6.12e-04	PASS
risk-neutral density mass	1	1	PASS
risk-neutral density mean = forward	1	1	PASS
density non-negativity (min/peak) tail dither scales as $1/dK^2$	-8.87e-10	-1.00e-07	PASS
sampling density: min $\geq \epsilon$ and mass = 1 floored 622 pts, removed -5.1e-10 of mass	0	1.00e-12	PASS

Estimators

check	measured	threshold	verdict
H from ATM skew (planted 0.12) $r^2 = 1.00000$	0.12	0.12	PASS
H from structure function, worst of 4 planted $H = 0.10, 0.12, 0.30, 0.50$	0.0062	0.03	PASS
two independent H routes agree surface route vs path route	0.002891	0.05	PASS
forward from put-call parity $r^2 = 1.000000$, no rate or dividend input	1.14e-13	1.00e-06	PASS
discount factor from parity	4.44e-16	1.00e-09	PASS
SVI parameter recovery	1.43e-08	0.001	PASS
SVI slice passes Durrleman $g_{\min}$ over data = 0.2297	1	1	PASS
butterfly audit: false positives on a convex slice	0	0	PASS
butterfly audit: catches a planted dent flags the dent's neighbours	2	1	PASS
calendar audit: clean on monotone w	0	0	PASS
calendar audit: catches a reversal	3	1	PASS

Calibration

Status

check	measured	threshold	verdict
plant-and-recover, worst relative error 7 iters, 0.6s	3.13e-11 rel	0.01 rel	PASS
fitted RMSE on clean data	4.06e-09 vol pts	0.001 vol pts	PASS
Jacobian condition number at the solution high = parameters trade off; see identifiability	30.05	1.00e+08	PASS
spread of v0 under 0.5vp noise stable	3.775 %	10 %	PASS
spread of kappa under 0.5vp noise unidentified	34.73 %	50 %	PASS
spread of theta under 0.5vp noise stable	2.803 %	50 %	PASS
spread of xi under 0.5vp noise loose	19.13 %	50 %	PASS
spread of rho under 0.5vp noise stable	3.73 %	10 %	PASS
spread of kappa with a Tikhonov prior (weight 50) prior from a history or a variance swap	0.07281 %	10 %	PASS

Markovian lift

check	measured	threshold	verdict
K(t) = $\int e^{-xt} \mu(dx)$ representation $\mu(dx) = x^{-a} / (\Gamma(a)\Gamma(1-a)) dx$, $a = H+1/2$	6.67e-10 rel	1.00e-08 rel	PASS
kernel error, N=24, t in [1 day, 2 y] max relative; H = 0.12	0.008697 rel	0.01 rel	PASS
kernel error on [1 hour, 2 y] a one-day option lives here; 12% below 5 minutes	0.008697 rel	0.01 rel	PASS
kernel error, N=24, worst over H in [0.08, 0.5] H = 0.05: 0.0105, H = 0.02: 0.0113 — above 1%	0.00972 rel	0.01 rel	PASS
kernel error, N=32, worst over the whole H box [0.02, 0.5] what calibration refits with when a fit lands below H = 0.08	0.006656 rel	0.01 rel	PASS
Laplace-domain error, z in [0.1, 100] (plan: 1%, see note) 3.5% is the memory beyond 2 y; 1% is not reachable with a 2 y fit	0.03508 rel	0.05 rel	PASS
Laplace-domain error on the pricing band z in [1, 100] a year down to a few days	0.01419 rel	0.03 rel	PASS
lifted recursion == SOE convolution (identity)	4.05e-14	1.00e-12	PASS
lifted path vs true-kernel Volterra path (L2, same noise)	0.008001 rel	0.01 rel	PASS
N=1 at x=0 reproduces closed-form Heston (ETDRK4, 200 steps)	7.09e-10	1.00e-09	PASS
full N-node lift at H = 0.4999 vs Heston shrinks 10x per decade of $(1/2 - H)$	1.64e-05	1.00e-04	PASS
ETDRK4 vs implicit trapezoidal + Richardson (H=0.12) independent time-steppers, 1 d / 30 d / 1 y	3.79e-06	5.00e-06	PASS
rough objective evaluation, 10 expiries x 13 strikes best of 3 (all: 0.97, 0.62, 0.60); trend target: rolling mean ≤ 0.75 s	0.6048 s	3 s	PASS

Rough robustness

check	measured	threshold	verdict
extrap at 120 steps, u to 1200, guards off (NOT unconditionally stable) $ \phi(u - i/2) $ must be ≤ 1 ; kept to prove the constraint	1.21e+36	1	PASS
stability constants enforced: the same solve raises StabilityError hard precondition since Session G; $ \phi_{ij} \leq 1$ is a hard postcondition	1	1	PASS
stability-sized extrap, same u range: max $ \phi(u - i/2) $ 383 steps	0.9988	1	PASS
runaway parameters: maturities priced in-band (of 4) Carr-Madan returned 1.9e18 / 2.9e28 at 30 / 90 d here	4	4	PASS
unpriceable quote: minimum charge (vol) vs 0 before exact fit costs 1.4e-22	4.754	4.754	PASS

Rough identifiability

Status

check	measured	threshold	verdict
SE(H), 20 quotes, equal weights, 0.5 vp noise	0.04711	0.06	PASS
SE(H), same quotes, vega^2 weights H not identified: the weights discard the short end	0.2024	0.15	PASS
corr(H, xi) at the truth the rough analogue of kappa/theta	0.9776	0.9	PASS

Hawkes

check	measured	threshold	verdict
mean rate vs $\mu/(1-n)$, in SE 12.457 vs 12.5	0.5936 SE	3 SE	PASS
count variance / Hawkes (1971) closed form, 0.25 y Fano 5.93	1.019	0.15	PASS
MLE plant-and-recover, worst z	1.004	3	PASS
time rescaling KS p, Hawkes fit (should pass)	0.5838	0.01	PASS
time rescaling KS p, Poisson fit to the same events clustering is detected, not assumed	0	1.00e-06	PASS
second spike: superposition identity, 4 hosts $\text{inc2} - \text{inc1} = r(d+W) - r(d)$	8.88e-16	1.00e-12	PASS
Heston: property switches on at $\alpha = \kappa$ (1.03 kappa) at 0.97 kappa: -2.1e-06	1.38e-06	0	PASS
rough (driver jumps): pairs with the property at $n = 0.6$ 14 of 21 at $n = 0.95$ — needs near-critical clustering	0	0	PASS
kurtosis vs scale-mixture closed form, in SE 5.73 vs 5.59; ratio to Poisson = daily Fano	0.7597 SE	4 SE	PASS

Kalman

check	measured	threshold	verdict
fast scalar path vs generic matrix filter	0	1.00e-12	PASS
RMSE improvement minus steady-state theory 45.6% vs 45.2%	0.3392 pts	2 pts	PASS
lecture AR(1) kappa on noisy quotes / true kappa attenuation bias; MLE through the filter is unbiased	12.6 x	2 x	PASS
dual vs joint EKF agreement	0.129 %	1 %	PASS
recursive MLE (Ljung) distance from the MLE Wan-Nelson dual: 0.07 SE	0.2287 SE	1 SE	PASS
strict: steps to re-converge after the regime change bad-print excursion 2.0x normal error	18	2	PASS
adaptive: bad-print excursion (x normal error) re-converges in 0 steps	14.92 x	6.044 x	PASS
robust: steps to re-converge (and ignores the print) excursion 1.7x vs strict 2.0x	1.5	2	PASS
lifted rough filter: RMSE improvement over quotes 24 lifted factors as the state, rank-one Q	39.67 %	30 %	PASS

Recording (real data)

check	measured	threshold	verdict
tau: 17:00 Istanbul to next close, hours a naive local clock gave 27 h (fixed in Session G)	30 h	30 h	PASS
US close in UTC follows daylight saving	1	1	PASS
delayed greeks (tick 83) recorded, vega per 1.00 vol IBKR sends vega per vol point	1 x	1 x	PASS

Positivity (Session G)

check	measured	threshold	verdict
QE step: mean vs exact conditional mean, in SE variance -0.27% vs exact; min V 2.7e-11	0.02983 SE	4 SE	PASS
quadrature of the integrated-variance moments	1.07e-15 rel	1.00e-12 rel	PASS
rough variance simulated 3000 days: minimum Euler at the same parameters: below zero on 13% of days	-2.08e-17	-1.00e-12	PASS
exact-moment filter: QML kappa bias away from zero, mean z near zero a Gaussian quasi-likelihood is unreliable (open flag)	0.6433 SE	2 SE	PASS

Lift fidelity (Session G)

Status

check	measured	threshold	verdict
7-day IV, lift N=24 vs true rough Heston (fractional Adams) 1 day: 0.15 vp, skew -1.6% (study_lewis_isometry.py)	0.0634 vp	0.1 vp	PASS
lto isometry of the lift at 1 day, shortfall vs true kernel 93% of it from lags under 7 minutes; prices integrate it away	21.6 %	25 %	PASS

H weighting (Session G)

check	measured	threshold	verdict
SE(H): inverse variance / vega^2 0.0749 -> 0.0076	0.1018 x	0.25 x	PASS
worst-case RMS error in H: hybrid (anchors x25) inverse variance 0.087, vega^2 0.098	0.03783	0.08654	PASS
min corr(H, xi) over all schemes and designs structural: re-weighting cannot remove it	0.9219	0.9	PASS

Learning H (Session G)

check	measured	threshold	verdict
filter profile likelihood: H_hat - 0.12 in SE H_hat 0.133 +/- 0.012, xi re-fitted at each H	1.051 SE	3 SE	PASS
log-likelihood of H = 0.49 against the best H Heston-like roughness rejected with xi free	-266	-20	PASS

Engineering

check	measured	threshold	verdict
scipy modules on the live import path scipy.stats alone costs 5.8 s	0	0	PASS
startup to actionable error, no TWS was 9.6 s before lazy imports	0.5423 s	3 s	PASS
calibration objective evaluation 39 quotes; 470 ms before vectorising	8.565 ms	200 ms	PASS
Gauss-Legendre rules constructed 2 reuses; leggauss is O(n^2)	1	1	PASS
quadrature tolerance 1e-12 vs 1e-14 calibration runs at 1e-12, ~2x faster	0	1.00e-12	PASS
replay round-trip fidelity recorded vs replayed surface points	0	1.00e-12	PASS

107 of 107 checks pass (58s).

The trend over the recorded runs (flags would appear in the last column):

Trend over 8 recorded run(s)

check	n	mean	std	last	flags
3-point stencil order on the same grid (kept f	8	1.086	0	1.086	
7-day IV, lift N=24 vs true rough Heston (frac	2	0.0634	0	0.0634	
Carr-Madan insensitive to damping alpha	8	6.11e-16	0	6.11e-16	
Carr-Madan vs exact Black-76	8	1.94e-16	0	1.94e-16	
ETDRK4 vs implicit trapezoidal + Richardson (H	8	3.79e-06	7.88e-18	3.79e-06	
Fourier vs Monte Carlo (sigmas)	8	0.3618	0	0.3618	
Gauss-Legendre rules constructed	8	1	0	1	
H from ATM skew (planted 0.12)	8	0.12	0	0.12	
H from structure function, worst of 4 planted	8	0.0062	0	0.0062	
Heston: property switches on at alpha = kappa	5	1.38e-06	0	1.38e-06	
Ito isometry of the lift at 1 day, shortfall v	2	21.6	0	21.6	
Jacobian condition number at the solution	8	30.05	0	30.05	
K(t) = int e^-xt mu(dx) representation	8	6.67e-10	0	6.67e-10	
Laplace-domain error on the pricing band z in	7	0.01419	3.75e-18	0.01419	
Laplace-domain error, z in [0.1, 100] (plan as	1	0.03508	0	0.03508	
Laplace-domain error, z in [0.1, 100] (plan: 1	7	0.03508	0	0.03508	
Lewis vs Carr-Madan on Heston	8	4.99e-15	0	4.99e-15	
Lewis vs exact Black-76	8	3.55e-15	0	3.55e-15	
MLE plant-and-recover, worst z	5	1.004	0	1.004	
N=1 at x=0 reproduces closed-form Heston (ETDR	8	7.09e-10	0	7.09e-10	
QE step: mean vs exact conditional mean, in SE	2	0.02983	0	0.02983	
RMSE improvement minus steady-state theory	5	0.3392	0	0.3392	
SE(H), 20 quotes, equal weights, 0.5 vp noise	6	0.04711	0	0.04711	
SE(H), same quotes, vega^2 weights	6	0.2024	0	0.2024	
SE(H): inverse variance / vega^2	2	0.1018	0	0.1018	
SVI parameter recovery	8	1.43e-08	0	1.43e-08	
SVI slice passes Durrleman	8	1	0	1	
US close in UTC follows daylight saving	2	1	0	1	
adaptive: bad-print excursion (x normal error)	5	14.92	0	14.92	
branch continuity, shipped (max/median step)	8	5.242	0	5.242	
branch discontinuity, reciprocal form	8	2825	0	2825	
butterfly audit: catches a planted dent	8	2	0	2	
butterfly audit: false positives on a convex s	8	0	0	0	
calendar audit: catches a reversal	8	3	0	3	
calendar audit: clean on monotone w	8	0	0	0	
calibration objective evaluation	8	9.711	2.347	8.565	
corr(H, xi) at the truth	6	0.9776	1.22e-16	0.9776	
count variance / Hawkes (1971) closed form, 0.	5	1.019	0	1.019	
delayed greeks (tick 83) recorded, vega per 1.	2	1	0	1	
density non-negativity (min/peak)	8	-8.87e-10	0	-8.87e-10	
discount factor from parity	8	4.44e-16	0	4.44e-16	
dual vs joint EKF agreement	5	0.129	0	0.129	
exact-moment filter: QML kappa bias away from	2	0.6433	0	0.6433	
exptrap at 120 steps, u to 1200 (NOT unconditi	4	1.21e+36	0	1.21e+36	
exptrap at 120 steps, u to 1200, guards off (N	2	1.21e+36	0	1.21e+36	
fast scalar path vs generic matrix filter	5	0	0	0	
fd_first exact on a quadratic	8	7.46e-14	0	7.46e-14	
fd_second on a quadratic vs its rounding floor	8	0.41	0	0.41	
filter profile likelihood: H_hat - 0.12 in S	2	1.051	0	1.051	
fitted RMSE on clean data	8	4.06e-09	0	4.06e-09	
forward from put-call parity	8	1.14e-13	0	1.14e-13	
full N-node lift at H = 0.4999 vs Heston	8	1.64e-05	2.33e-17	1.64e-05	
implied-vol round trip	8	8.05e-16	0	8.05e-16	
kernel error on [1 hour, 2 y]	8	0.008697	0	0.008697	
kernel error, N=24, t in [1 day, 2 y]	8	0.008697	0	0.008697	
kernel error, N=24, worst over H in [0.08, 0.5	2	0.00972	0	0.00972	
kernel error, N=32, worst over the whole H box	2	0.006656	0	0.006656	
kurtosis vs scale-mixture closed form, in SE	5	0.7597	0	0.7597	
lecture AR(1) kappa on noisy quotes / true kap	5	12.6	0	12.6	
lifted path vs true-kernel Volterra path (L2,	8	0.008001	0	0.008001	
lifted recursion == SOE convolution (identity)	8	4.05e-14	0	4.05e-14	
lifted rough filter: RMSE improvement over quo	5	44.31	4.242	39.67	
log-likelihood of H = 0.49 against the best H	2	-266	0	-266	
mean rate vs mu/(1-n), in SE	5	0.5936	0	0.5936	
min corr(H, xi) over all schemes and designs	2	0.9219	0	0.9219	

Status

naive-form cancellation at xi=1e-6 (kept to pr	8	3.44e-05	0	3.44e-05
norm_cdf vs scipy	8	2.22e-16	0	2.22e-16
norm_pdf vs scipy	8	0	0	0
phi(-i) = 1 (martingale)	8	0	0	0
phi(-u) = conj(phi(u))	8	0	0	0
phi(0) = 1	8	0	0	0
plant-and-recover, worst relative error	8	3.13e-11	0	3.13e-11
put-call parity C-P = F-K	8	2.84e-14	0	2.84e-14
quadrature of the integrated-variance moments	2	1.07e-15	0	1.07e-15
quadrature tolerance 1e-12 vs 1e-14	8	0	0	0
recursive MLE (Ljung) distance from the MLE	5	0.2287	0	0.2287
replay round-trip fidelity	8	0	0	0
risk-neutral density mass	8	1	0	1
risk-neutral density mean = forward	8	1	0	1
robust: steps to re-converge (and ignores the	5	1.5	0	1.5
rough (driver jumps): pairs with the property	5	0	0	0
rough objective evaluation, 10 expiries x 13 s	8	0.7122	0.16	0.6048
rough variance simulated 3000 days: minimum	2	-2.08e-17	0	-2.08e-17
runaway parameters: maturities priced in-band	6	4	0	4
sampling density: min >= eps and mass = 1	8	0	0	0
scipy modules on the live import path	8	0	0	0
second spike: superposition identity, 4 hosts	5	8.88e-16	0	8.88e-16
shipped 2nd derivative: order on a \$1/\$5 kink	8	3.077	0	3.077
simulated E[S/F] = 1	8	1.94e-06	0	1.94e-06
spread of kappa under 0.5vp noise	8	34.73	0	34.73
spread of kappa with a Tikhonov prior (weight	8	0.07281	0	0.07281
spread of rho under 0.5vp noise	8	3.73	0	3.73
spread of theta under 0.5vp noise	8	2.803	0	2.803
spread of v0 under 0.5vp noise	8	3.775	0	3.775
spread of xi under 0.5vp noise	8	19.13	0	19.13
stability constants enforced: the same solve r	2	1	0	1
stability-sized exptrap, same u range: max ph	6	0.9988	0	0.9988
startup to actionable error, no TWS	8	0.602	0.1361	0.5423
strict: steps to re-converge after the regime	5	18	0	18
tau: 17:00 Istanbul to next close, hours	2	30	0	30
time rescaling KS p, Hawkes fit (should pass)	5	0.5838	0	0.5838
time rescaling KS p, Poisson fit to the same e	5	0	0	0
trap / shipped separation	8	538.9	0	538.9
two independent H routes agree	8	0.002891	0	0.002891
unpriceable quote: minimum charge (vol) vs 0 b	6	4.754	0	4.754
vega vs central difference	8	4.23e-09	0	4.23e-09
worst-case RMS error in H: hybrid (anchors x25	2	0.03783	0	0.03783
xi -> 0 degenerates to Black-Scholes	8	2.22e-16	0	2.22e-16
phi(u) <= 1	8	0.9999	0	0.9999
0 flagged				